

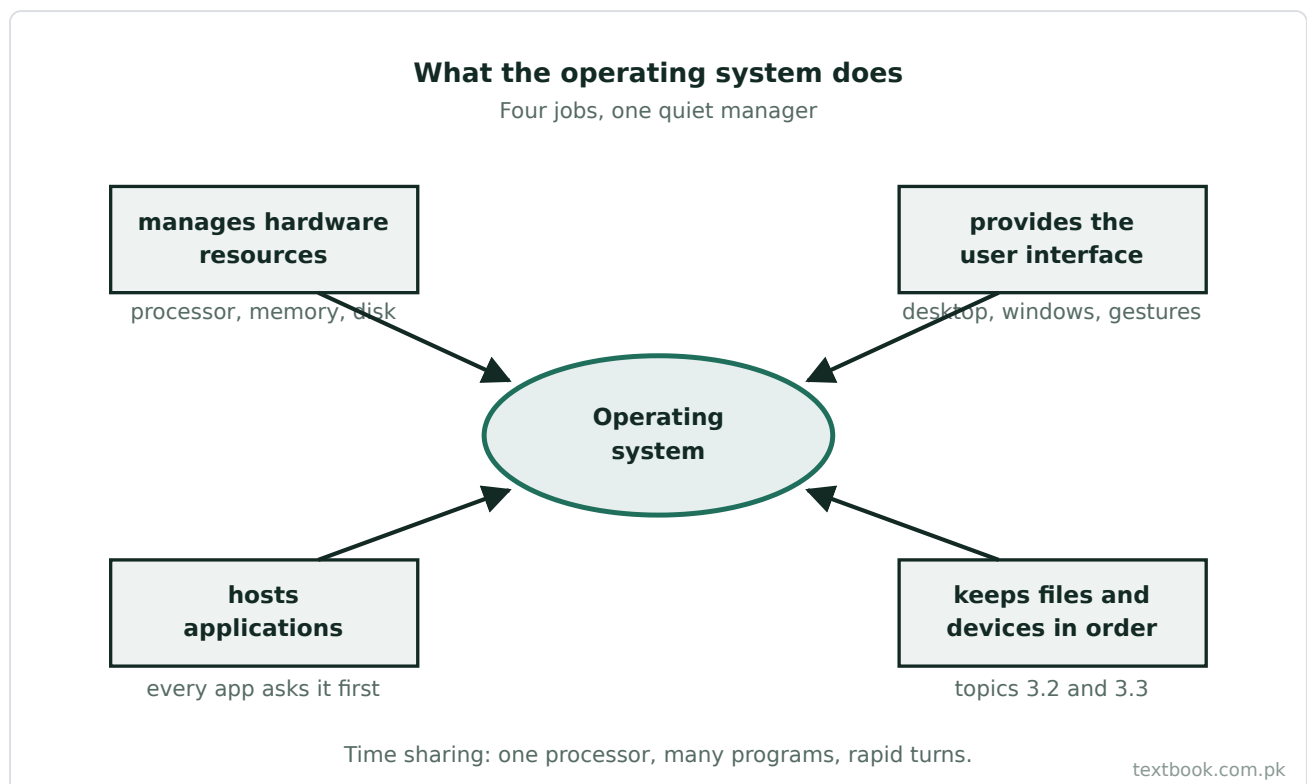
Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 3.1 - What an operating system does

A real classroom situation

The examination week at a Hyderabad college runs on one shared office computer. While the clerk prints admit cards, the head teacher's video call is running, the results file is open, a USB drive is being scanned for viruses, and a Windows update notification keeps appearing in the corner. Nothing crashes. One machine, five things happening at once, one piece of software calmly dividing attention: the operating system. The clerk never thinks about it, which is exactly the measure of how well it works.



Explanation

Functions of an operating system

From Unit 2 you already know the operating system's three-part job description: managing the hardware resources of the computer, providing the user-interface components, and providing a platform for software developers to write applications (Bourgeois et al., 2019). This topic widens that description into the day-to-day functions.

The OS **manages resources**. One processor, one pool of memory, one disk, one network connection - yet many programs want them all. The operating system divides and guards: it decides which program runs next and for how long, hands out memory, and polices the boundary so one program cannot walk over another's work. Arpaci-Dusseau and Arpaci-Dusseau (2023) describe the underlying trick as virtualisation: by running one process, then stopping it and running another, the OS creates the illusion of many processors when there is only one, a technique known as time sharing.

The OS **provides the interface**. The desktop, the windows, the touch gestures, the file icons: all drawn and managed by the operating system. It also provides the command line for those who prefer words to pictures.

The OS **hosts applications**. Every application you open asks the operating system for its share of memory, screen and processor time. The OS is the platform on which every other program stands, which is why an application written for one operating system does not automatically run on another.

And the OS **keeps files and devices in order**: it maintains the file system you meet in Topic 3.2 and manages the input/output traffic you meet in Topic 3.3. When you plug in a USB drive and a window appears offering to open it, that is the operating system noticing a device and asking what you want.

Five types of operating system

The same functions, tuned to different promises

Type	How it works	One example use
Batch	jobs collected and run one after another without user interaction	early payroll processing
Time-sharing	many programs or users share the processor in quick turns	a college office computer running five tasks
Real-time	must respond within a strict time limit	a patient monitor or an airbag controller
Distributed	one system spread across several connected machines	a university records system on many servers
Mobile	designed for touch phones with a small screen and limited battery	Android on a school tablet
Promises: fairness (time-sharing), speed (real-time), scale (distributed), endurance (mobile).		

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Types of operating systems

Operating systems come in types, shaped by the job. A **batch** operating system runs collected jobs one after another without user interaction - the historical pattern of early business computing, where a day's payroll was submitted and processed as a batch. A **time-sharing** (multi-tasking) system shares the processor among many programs or users in quick turns, which is every desktop and laptop you have used: the office computer juggling printing, a call and a spreadsheet is time sharing at work. A **real-time** system must respond within a strict time limit; a patient monitor that must raise its alarm within milliseconds, or a car's engine controller, cannot wait for a convenient turn. A **distributed** system spreads one operating system's work across several connected machines, as a university's records system may run on several servers at once. And a **mobile** operating system, such as Android or iOS, is designed around touch, a small screen and a battery that must last the day.

The types are not rival species; they are the same set of functions tuned for different promises. The office computer promises fairness (everyone gets a turn); the monitor promises speed (the turn cannot wait); the phone promises endurance (the battery must survive the day).

A caution against the most common misconception: the operating system is not just the desktop picture and the taskbar. Those are its interface, its face. The work - dividing the processor, guarding memory, keeping the file list, feeding devices - happens continuously underneath, and the next two topics look straight at that work.

Activity: Function spotting

Work in pairs. About 20 minutes.

1. Watch your teacher perform five actions at one computer: open two applications together, copy a file to a USB drive, unplug the drive without asking, print a page, and leave the computer idle for a minute.
2. For each action, name which operating system function is on display (resource management, interface, hosting applications, file management, input/output management) and write one sentence describing what the OS did.
3. Then classify the classroom computer's operating system by type, and say which promise (fairness, speed, endurance) matters most for that machine and why.
4. Compare answers with another pair and settle any disagreement by pointing to what happened on screen.

Check your understanding

1. Name the three-part job description of an operating system from Unit 2.
(*Remember*)
2. Explain time sharing in two sentences, using the idea of the illusion of many processors. (*Understand*)
3. A hospital's patient-monitoring machine and a school office desktop run different operating system types. Name each type and explain the promise each makes. (*Apply*)
4. Why is it wrong to say the operating system is "just the desktop picture"?
(*Understand*)

Summary

The operating system manages resources (dividing the processor by time sharing and guarding memory), provides the user interface, hosts every application, and keeps files and devices in order. Its types - batch, time-sharing, real-time, distributed, mobile - are the same functions tuned to different promises: fairness, strict response time, scale or battery endurance. The desktop picture is its face, not its work.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the operating system's functions and give an everyday example of each.
- ☐ I can explain time sharing in my own words.
- ☐ I can name the five OS types and match a described use case to one of them.

Try this at your practicum school

Ask to watch the office computer for ten minutes during a busy spell. Note every visible sign of the operating system at work (windows switching, a print dialog, a USB notification, a password prompt). Write down which function each sign belongs to.

Summative task

The head teacher wants to buy a machine to run the school's biometric attendance system, which must record every attendance check instantly and never wait. Using the types of operating system from this topic, recommend which type the machine's system should resemble and why, and explain what would go wrong if an ordinary time-sharing desktop system were used instead.

Marking guide (10 marks):

- The correct OS type identified with its defining promise: 3 marks.
- The recommendation justified against the requirement of instant response (Analyse): 4 marks.
- The failure mode of the time-sharing alternative explained concretely: 2 marks.
- Clear writing: 1 mark.

Further reading

- Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2023). *Operating Systems: Three Easy Pieces* (version 1.10), chapter 4.
<https://pages.cs.wisc.edu/~remzi/OSTEP/>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 3. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Norton, P. *Introduction to Computers*. McGraw-Hill Education. (Course guide reading; print monograph, cited at bibliographic level.)

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